Avoiding the British: Brexit and Partner Choice in European Venture Capital

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Abstract

Nothing in the United Kingdom's withdrawal from the European Union restricted co-investment. No rule, before or after, required a continental venture capitalist to seek authorisation before syndicating a round with a British one. Continental investors whose pre-determined reliance on British partners was above the median nonetheless reduced the share of their syndicates including a British investor by 8.2 percentage points against a base rate of 18.7% ($p < 0.001$; 95% CI $[-13.1, -3.2]$), a relative decline of forty-four percent. The decline begins in the semester of the referendum and persists through 2025, five years after the only legal change that ever occurred. Using Crunchbase and an exposure-based design with wild-cluster bootstrap inference, we show that the effect is confined to the relational margin: deal counts, stages, target geography, round sizes, follow-on financing and exits are all null, established British ties dissolve no faster than comparable continental ones (triple difference +0.3 pp, $p = 0.77$), and the aggregate volume of UK–continental syndication tracks the rest of Europe within $[-20\%, +26\%]$. What moved was whom these investors chose as partners, and nothing else. Decomposing that choice, the effect is identified on the formation of new British ties, the only margin, and the British corridor the only one of thirteen, to survive the falsification protocol, and not on the continuation of existing ones. That narrows the admissible channels without settling among them. That a shock which changed no rule governing syndication nonetheless reallocated it suggests that European financial integration rests on behaviour at least as much as on the legal framework usually credited with sustaining it.

1 Introduction

Venture capital deals are syndicated, and syndication is relational. A co-investment tie is not a spot transaction but an asset: its value is the flow of deals the partnership is expected to generate. Before June 2016, British venture capitalists were among the most sought-after partners in Europe, combining reputation, capital, and access to the deepest European market. The referendum of 23 June 2016 was substantially unanticipated by financial markets (Breinlich et al., 2018); sterling fell eight percent overnight and UK policy uncertainty reached a record (Baker et al., 2016).

It changed nothing about the legal capacity of a French fund to write a cheque alongside a British one. That capacity was never conferred by Union membership, was never passported, and was never withdrawn. A continental venture capitalist could syndicate with a British venture capitalist on 22 June 2016, on 24 June 2016, on 1 February 2020, and today, under identical terms and with identical formalities.

This paper documents that they stopped doing so anyway.

Among continental venture capitalists that all had at least one British co-investor before the vote, those whose pre-determined reliance on UK partners was above the median durably reduced the share of their syndicates including a British investor by 8.2 percentage points ($p < 0.001$; 95% CI $[-13.1, -3.2]$) against a control base rate of 18.7%, a relative decline of forty-four percent. The decline begins in the semester containing the referendum, deepens through the withdrawal and the end of the transition period, and is still present in 2022–2025. It is present in each of three post-referendum windows: -6.4 points in H2 2016–2019, -10.6 in 2020–21, -8.3 in 2022–25.

Timing matters here more than usual. The United Kingdom remained inside the single market until 31 December 2020. If the mechanism were legal, the response should follow the law. It precedes it by four and a half years and does not accelerate when the law finally changes.

What makes the shift worth a paper is not its size but how narrowly it is confined. We look for it in every adjacent margin and do not find it.

It did not break relationships. We construct the universe of investor dyads established before 2014 and ask whether a persistent UK–continental pair is more likely to dissolve after the referendum than a persistent continental–continental pair. The triple difference is $+0.3$ percentage points ($p = 0.77$; 95% CI $[-1.8, +2.5]$ on a semi-annual reactivation base rate of 5.6%). Persistent ties decay everywhere, by 2.3 points on average, and they decay at the same rate whether the partner sits in London or in Milan.

It did not shrink the corridor. Comparing the eighteen UK–continental country-pair corridors against fifty-five continental–continental corridors, the post-referendum differential in the volume of jointly syndicated deals is $+0.03$ log points ($p = 0.81$),

with a confidence interval that bounds any effect within $[-20\%, +26\%]$.

It did not change what these investors did. Deal count, stage composition, the geography of their targets, the size of the rounds they join, the probability that their portfolio companies raise again or exit within four years: all null, all estimated on the same sample with the same design (Appendix B). The apparent exception, an increase in activity among small exposed investors, does not survive falsification: exposure to American, Swiss or non-EU OECD partners produces the same coefficient or larger, and the effect is absent from the window immediately following the referendum.

The effect is therefore located on one margin and one only: the choice of partners. These investors did not invest less, elsewhere, differently, or worse. They stopped picking the British.

A semester in which one of these investors syndicates with a British partner is of exactly one of two kinds: either a British partner in it is being met for the first time, or every British partner in it has been met before. The two indicators exhaust the outcome, so the same specification run on each splits the headline exactly: -4.8 points on the formation margin, -3.4 on the continuation margin. What separates them is not their size but the falsification protocol. On the formation margin the British corridor is the only one of the thirteen to survive it: clean pre-trends on both joint tests, a significant static coefficient, and an effect of consistent sign in each of the three post-referendum windows. On the continuation margin no corridor survives it, the United Kingdom included.

The margin narrows the field. An anticipated bilateral friction attaches to the partner and not to the acquaintance, and a tightening of European public limited partners' geographic mandates would bind on any new investment, a follow-on alongside a British investor already in the syndicate included. Both predict a response on the two margins; the protocol identifies one on a single margin. What remains are the channels that operate where partners are matched: the coordination cost of entering a new bilateral relationship, and the salience of a redrawn institutional boundary. Our data cannot separate those two, and we do not assert one. Section 8 reports the split and what each channel would require.

To arrive at these conclusions, we build four panels from Crunchbase, each designed so that treated and control units are subject to the same shock and differ only in pre-determined exposure. The first is an investor panel: 448 continental VCs that all had at least one British co-investor before 2014 (216 above the median exposure share, 232 below), observed over 23 semesters from 2014 to 2025. Because a single British tie mechanically pushes a small investor's exposure share above the median, every specification carries a small $\times$ post control that nets out the resulting size trend. The second is a dyad panel: investor pairs established before 2014 (1,347 with a British partner and 7,224 purely continental), observed over 197,133 dyad-semesters, on which we estimate a triple difference. The third is a corridor panel: 73 country-pair corridors, of which 18 link the United Kingdom to the continent

and 55 link two continental countries, with the outcome defined identically for both. The fourth mirrors the first on the other side of the Channel: 255 British VCs that all had at least one continental co-investor before 2014. Exposure is classified using only deals concluded before 2014, so that the classification cannot be contaminated by anticipation, and the semesters 2014–2015 are held out to test whether the groups were parallel before the shock. Inference is by wild-cluster bootstrap-t throughout, with between twenty-one and seventy-three clusters depending on the panel; confidence intervals are obtained by inverting the bootstrap distribution, and joint pre-trend tests are computed by restricted wild-cluster bootstrap rather than by the asymptotic Wald statistic, which over-rejects at these cluster counts.

We establish a compositional reallocation of European venture syndicates away from British partners, date it to the referendum rather than to the legal separation, and bound its extent on every adjacent margin we can measure. We show it is specific: submitted to an identical, pre-specified protocol, twelve placebo corridors do not reproduce it on the relevant date. And we draw the implication we believe is the most portable: legal access and observed integration are distinct objects. The rules governing syndication were intact throughout the response, and the pattern of partner choice changed anyway.

Table 1 collects the results.

Table 1: The effect and its boundaries.

	Outcome	Estimate	95% CI	Verdict
§6	UK-partner rate (pp)	-8.20	$[-13.11, -3.15]$	**Identified**
§8	of which, formation margin (pp)	-4.84	$[-9.24, -0.18]$	**Identified**
§8	of which, continuation margin (pp)	-3.36	$[-7.16, +0.24]$	Not identified
§7.1	Dyad survival, triple diff. (pp)	$+0.30$	$[-1.84, +2.51]$	Precise null
§7.2	Corridor volume (log pts)	$+0.03$	$[-0.20, +0.26]$	Precise null
§7.3	Activity, window $[0, 6]$ (pp)	$+3.18$	$[-1.47, +8.16]$	Not identified
§7.4	British side, first stage (pp)	-6.01	$[-11.52, -0.33]$	Not identified

Notes. Wild-cluster bootstrap-t intervals throughout (Appendix A reports base rates and minimum detectable effects). The two precise nulls exclude economically meaningful effects: no reduction in the survival of British ties beyond a third of the baseline rate, and no change in corridor volume outside $[-20\%, +26\%]$. "Not identified" is not "zero": the last row's interval excludes zero, and the distinction is discussed in Section 7.4.

Section 2 establishes that no rule governing syndication changed. Section 3 situates the paper. Section 4 presents the data and Section 5 the design. Section 6 presents the effect, Section 7 its boundaries, Section 8 the margin on which the shock operates, and Section 9 the implications.

2 A shock that never regulated syndication

On 23 June 2016, 52% of UK voters chose to leave the European Union. The outcome was not priced: sterling fell eight percent overnight and the UK Economic Policy Uncertainty Index reached an all-time high (Baker et al., 2016; Breinlich et al., 2018). Legal separation followed much later: withdrawal on 31 January 2020, end of the transition period on 31 December 2020.

The central feature of this setting, and the one that makes the finding worth reporting, is that none of these dates altered the rules governing co-investment.

Syndication is not a passported activity. The acquisition of an equity stake in a private company by a professional investor is not conditioned on the investor's country of establishment being a member of the Union. Two funds may invest in the same round without notifying any authority of their joint presence, without a common licence, and without any instrument that Union membership confers. The constraint most often invoked in this context, the loss of the marketing passport conferred by the Alternative Investment Fund Managers Directive, governs the *management and marketing of funds to professional investors*, that is, the raising of capital from limited partners. It does not govern the deployment of that capital into a portfolio company, and it remained in force for the United Kingdom until the end of 2020 in any case. Whatever the passport did, it did not make a British co-investor harder to add to a syndicate.

This has three consequences for what follows.

First, a mechanical explanation is unavailable. We cannot attribute the shift to a rule that required it, because there is no such rule, at any date in our sample.

Second, timing becomes informative rather than incidental. If the response were driven by legal separation, it should appear at withdrawal or at the end of transition. It appears at the referendum, and Section 6 shows it is already established in the window running from H2 2016 to 2019, entirely inside the single market period.

Third, the shock is composite (regulatory, macroeconomic, political), and we make no attempt to decompose it. Our object is the effect of the referendum on partner choice, not the relative weight of the channels through which it travelled, which our data cannot identify. We estimate what happened; we do not assert a mechanism we cannot test.

3 Related literature

3.1 Syndication as relational capital

Venture capital deals are overwhelmingly syndicated, and the literature has established why. Lerner (1994) first documented that syndication is not merely risk-sharing: experienced investors syndicate with one another at the first round, which is consistent with a screening motive. Brander et al. (2002) formalise the tension between the two classical rationales, improved venture selection through a second opinion versus the value added by complementary skills, and find in favour of the latter. Casamatta and Haritchabalet (2007) extend this by showing that syndication is not costless to the junior partner. Manigart et al. (2006) document the determinants of the syndication decision across European markets.

What makes syndication ties an object of study in their own right is their persistence. Sorenson and Stuart (2001) show that syndication networks propagate opportunities across geographic space, so that an investor's position in the network determines what it gets to see. Hochberg et al. (2007) demonstrate that better-networked venture capitalists enjoy significantly better fund performance, and Hochberg et al. (2010) that incumbent networks constitute a barrier to entry: a tie is a durable, rivalrous asset. Partner choice is therefore not a detail of execution but the mechanism by which an investor's future opportunity set is determined, which is why a shift in whom investors select is economically consequential even when volumes and relationships are unchanged.

A parallel strand studies what investors bring to the companies they back, and thereby what a portfolio company loses when the composition of its syndicate changes. Hellmann and Puri (2002) show that venture capitalists professionalise their portfolio companies; Gompers (1995) that they monitor through staging; Bernstein et al. (2016) that this monitoring is causal. Reputation carries value that transfers to the company: Megginson and Weiss (1991) and Stuart et al. (1999) establish the certification effect at IPO, Hsu (2004) shows that entrepreneurs pay for affiliation with high-reputation investors, and Nahata (2008) that investor reputation predicts portfolio performance. If British investors were disproportionately providing certification to continental syndicates, their withdrawal should have degraded downstream outcomes. Appendix B shows it did not.

3.2 Cross-border venture capital and its frictions

Venture capital is stubbornly local. Cumming and Dai (2010) document a pronounced local bias in investment; Sorenson and Stuart (2001) attribute part of it to the structure of the syndication network itself. Cross-border investment therefore requires a mechanism to overcome distance, and syndication with a local partner is the canonical

one: Tykvová and Schertler (2012) show that syndicating with a local VC moderates the liability of foreignness. Schertler and Tykvová (2012) find institutional and informational determinants of cross-border inflows to dominate mechanical ones.

The frictions are not only informational but cultural and institutional. Guiso et al. (2009) show that bilateral trust predicts bilateral trade and portfolio investment; Bottazzi et al. (2016) demonstrate that the same holds within European venture capital specifically. This literature furnishes the strongest prior for what we find: if a political rupture degrades bilateral trust, and trust supports cross-border venture investment, then partner choice is exactly the margin on which a referendum should bite, and volumes and contracts need not move at all.

3.3 Brexit and the financial sector

The macroeconomic and trade consequences of the referendum are well documented. Baker et al. (2016) show that UK policy uncertainty reached an all-time high; Breinlich et al. (2018) extract the market's revision of expectations from the stock-price response; Dhingra et al. (2017) quantify the trade effects; Steinberg (2019) models the macroeconomic cost of trade policy uncertainty; Born et al. (2019) estimate the output cost using synthetic control methods.

Within venture capital, the closest paper is Bosio et al. (2025), published in *Research Policy* and co-authored with the research department of the European Investment Fund. Using the European Data Cooperative at the level of functional urban areas, they document that UK hubs cut their cross-border investment immediately after the announcement while continental hubs increased theirs after enforcement. Their object is flows and geography at the level of the urban hub; ours is partner choice at the level of the firm, identified off pre-determined exposure. The two are complementary.

3.4 Inference

Our inference follows the literature on clustering with a small number of groups. Cameron et al. (2008) introduce the wild-cluster bootstrap; MacKinnon and Webb (2017) characterise its behaviour under heterogeneous cluster sizes, and MacKinnon et al. (2023) provide the current guide to practice. With between twenty-one and thirty country clusters, asymptotic cluster-robust inference substantially over-rejects. This applies to joint pre-trend tests as much as to coefficients, a point we found to matter materially: the asymptotic Wald statistic rejects parallel pre-trends in specifications where no individual pre-period coefficient has a t-statistic above 1.2. We therefore compute pre-trend tests by *restricted* wild-cluster bootstrap, imposing the null in the bootstrap data-generating process, following the recommendation of MacKinnon and Webb (2017) for joint tests. On designs with many clusters the two procedures agree; on designs with few they do not, and the asymptotic version is the

one that misleads.

Finally, several of our results are nulls, which raises a question about why such papers are rare. Brodeur et al. (2016) document the discontinuity in the distribution of published test statistics around conventional thresholds; Andrews and Kasy (2019) formalise the resulting selection. A null is informative when it is precise and when it bounds a claim that has been made. Appendix A reports what each of ours excludes, and we say explicitly which are precise and which are merely underpowered.

4 Data

We use Crunchbase, from which we build an investment-level panel linking each funding round to each participating investor, with the country of the investor's headquarters and of the portfolio company. We restrict rounds to venture types (pre-seed through Series J, angel, and unspecified venture) and to OECD-domiciled targets. The resulting panel contains 1,211,489 investor–round observations.

The unit of time is the semester. We write $\text{sem} = 2 \times \text{year} + \mathbb{1}\{\text{month} > 6\}$ and define τ, event time, relative to 2016H2, the first semester wholly after the referendum. Thus $\tau = 0$ is 2016H2, $\tau = -1$ is the referendum semester itself (which we never use as reference), and $\tau = -2$ is 2015H2, our omitted category.

"Continental Europe" (CE) comprises the European countries excluding the United Kingdom and Ireland. A round is UK-exposed if at least one participating investor is headquartered in the United Kingdom.

Ireland is the only country we remove by name, so it owes the reader a sentence. Irish investors do not stand towards British ones as the rest of the continent does: the land border and the Common Travel Area survived the referendum intact, and Dublin was itself a destination for financial entities leaving London after it. That reason was fixed before we looked, which makes it exactly the kind of reason this paper distrusts elsewhere. We therefore test it rather than rest on it. Adding Ireland brings the universe from 448 investors to 470 and moves the estimate from -8.20 points to -8.01 ($p < 0.001$; 95% CI $[-12.62, -3.26]$), with both pre-trend tests still passing. A reader who prefers the wider definition may read -8.0 for -8.2 and nothing else follows. Ireland cannot be a corridor in its own right: twenty-two eligible investors return -6.8 points with an interval of $[-49.8, +29.7]$, eighty points wide, far below the minimum cell that Section 5 requires. We have run this check on the main estimate only.

Table 2: Summary statistics: the estimation sample.

	High exposure	Low exposure
Investors	216	232
Pre-2014 deals (mean)	10.51	32.48
Pre-2014 deals with a UK partner (mean)	3.59	2.91
UK exposure share (mean)	0.429	0.108
Small (≤ 5 pre-2014 deals), %	58.8	5.2

Notes. Continental VCs with at least two pre-2014 deals and at least one pre-2014 British co-investor. "High" denotes an exposure share above the sample median.

The two groups differ in size: highly exposed investors are, mechanically, smaller, because a given number of British partnerships represents a larger share of a smaller book. We do not regard this as a threat, since investor fixed effects absorb the level, but any outcome correlated with size and trending over the sample would contaminate the estimate. We therefore include a $\text{small}_i \times \text{post}_s$ control in every specification.

Crunchbase is assembled from public announcements, and coverage follows visibility rather than a sampling frame. Coverage is thinner at the earliest stages, thinner outside the large western markets, and it records only investors identified as organisations with a headquarters country. Our sample is continental VCs with at least two pre-2014 deals and a pre-2014 British partner: firms visible enough to be recorded at least three times. The result therefore describes the behaviour of established, observable European investors, and we make no claim about the invisible tail. If the effect on syndication were concentrated among small, informal, or peripheral investors, our design would not find it.

It records the investors in a round, not the limited partners behind those investors. We therefore cannot observe whether a continental fund is backed by a European public institution, which is the one observable that would allow us to test the upstream channel discussed in Section 8. We flag this as a limitation of the source rather than of the design: the test is feasible with data on fund-level LP composition, and we do not have it.

5 Design

5.1 No untreated unit

Brexit reaches every European venture capitalist on the same date. There is no group the referendum did not touch. A conventional difference-in-differences is therefore unavailable, and a simple pre/post comparison is uninterpretable: it cannot distinguish the referendum from anything else that happened to European venture capital after

June 2016, and a great deal did, culminating in the global venture boom of 2020–2021.

Our identification comes from *differential exposure*. Among agents all treated by the same shock, those whose pre-determined reliance on the affected channel was higher should respond more. Semester fixed effects absorb the common component; the interaction of exposure with event time identifies the relative response. The estimand is a *relative* effect, and we are explicit that it is: we cannot recover the level effect of Brexit on European venture capital, and no design on these data can.

5.2 Specification

For investor i in semester s,

$$y_{is} \;=\; \sum_{\tau \neq -2} \beta_\tau \cdot \mathbb{1}\{s = \tau\} \cdot \mathrm{High}_i \;+\; \alpha_i \;+\; \delta_s \;+\; \varepsilon_{is}, \tag{1}$$

with investor and semester fixed effects. High_i is a pre-determined exposure indicator. Standard errors are clustered on the investor's country, of which there are between twenty-one and thirty depending on the sample. We report wild-cluster bootstrap-t p-values throughout (Rademacher weights, equal-tailed, $B = 999$), and we invert the bootstrap distribution to obtain confidence intervals. Joint pre-trend tests are computed by restricted wild-cluster bootstrap, as described in Section 3.

5.3 Early classification

Classifying exposure on the full pre-referendum window invites regression to the mean: an investor observed with an unusually high share of UK deals in 2015 will mechanically revert. We therefore classify exposure using *only* deals concluded before 1 January 2014, well before any Brexit signal, and begin the measurement panel in 2014. This leaves $\tau \in \{-5, -4, -3\}$ as a held-out window, used exclusively to test whether the groups were parallel before the shock arrived. Section 7.1 shows this correction is not cosmetic: it is the difference between a spurious -7.1 point "divorce" and a true zero.

5.4 The falsification protocol

We fix, before looking at any result, a single mechanical protocol applied identically to the United Kingdom and to twelve placebo partner corridors. For each corridor we compute: (i) the static exposure-by-post coefficient; (ii) a joint pre-trend test on $\tau \in \{-5, -4, -3\}$; (iii) a stricter joint pre-trend test additionally including $\tau = -1$, the referendum semester itself; and (iv) three post-referendum windows: $\tau \in [0, 6]$ (H2 2016 to 2019), $\tau \in [7, 11]$ (2020–21), and $\tau \in [12, 17]$ (2022–25).

The reading grid, also fixed ex ante: a genuine response requires a significant static coefficient, clean pre-trends on *both* tests, and an effect of consistent sign across all three windows including the last. An effect confined to 2020–21 is a pandemic-and-boom phenomenon. An effect absent from the first window is not a referendum effect either. The United Kingdom is subjected to exactly the same grid as the placebos, and we report the verdict the grid returns, whatever it is.

6 The shift in partner choice

The universe is continental VCs with at least two pre-2014 deals and at least one pre-2014 British co-investor. Every investor in this sample is UK-exposed; the contrast is between those above and below the median exposure share. The outcome huk_{is} equals one if investor i syndicated with a British investor in semester s.

Figure 1 plots Equation (1). The decline begins in the semester containing the referendum and does not reverse over the following nine years.

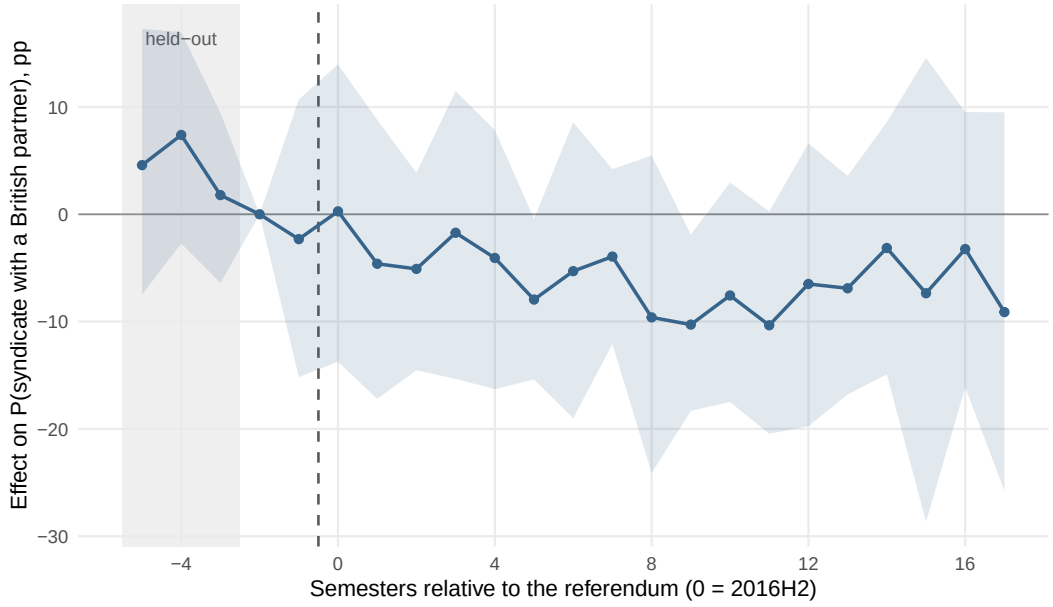


Figure 1: Effect on the probability that a continental VC syndicates with a British partner in a given semester, high versus low pre-2014 exposure. Early classification, investor and semester fixed effects, wild-cluster bootstrap-t bands ($B = 999$, clustered on investor country). Reference semester: 2015H2. The shaded band on the left is the held-out window used to test parallel pre-trends.

Two features of the figure deserve comment, and we prefer to raise them ourselves. The semester-by-semester coefficients are imprecise: only two of the eighteen post-referendum estimates have a bootstrap interval excluding zero. This is a power prob-

lem, not an identification problem (each semester is a single draw from 448 investors), and the effect is identified off the post-period average, which is precisely estimated.

More seriously, the held-out coefficients are not flat. Relative to 2015H2, the high-exposure group sits 4.6 and 7.4 points *above* the low-exposure group in 2014, and the gap closes monotonically thereafter. The joint pre-trend tests pass ($p = 0.52$ on $\tau \in \{-5, -4, -3\}$ and $p = 0.70$ when the referendum semester is added), but as Roth et al. (2023) emphasises, such tests have low power, and passing one is weaker evidence than the phrase "parallel pre-trends" usually conveys. We therefore state the limitation plainly: we cannot exclude that part of the post-referendum decline continues a downward drift already under way. What argues against that reading is the placebo evidence rather than the pre-trend test.

The pooled result is in Table 3. Highly exposed continental VCs reduce their UK-partner rate by 8.2 percentage points relative to less exposed peers ($p < 0.001$; 95% CI $[-13.1, -3.2]$), against a control base rate of 18.7%: a relative decline of forty-four percent. The effect is present in all three windows: -6.4 points in H2 2016–2019, -10.6 in 2020–21, -8.3 in 2022–25.

Table 3: The thirteen-corridor battery. Identical protocol, fixed ex ante.

Partner	Static	Pre-trends		Windows		Verdict
	(pp)	(i)	(ii)	$[0, 6]$	$[12, 17]$	
GBR	$\mathbf{-8.2}^{***}$	0.52	0.70	-6.4^{**}	-8.3^{**}	**Clears the grid**
CHE	-9.2^{***}	0.98	0.83	-10.9^{***}	-9.2^{***}	Clears the grid
USA	-2.1	0.18	0.31	-0.3	-2.7	null
CAN	$+0.5$	0.28	0.34	-2.1	$+4.2$	null
ISR	-0.7	0.52	0.09	$+0.8$	-1.3	null
JPN	-0.8	0.78	0.64	$+0.3$	$+1.8$	null
KOR	-4.6	0.37	0.06	-3.0	-4.0	null
FRA	-3.0	0.56	0.68	-1.3	-5.0	null
DEU	-3.0^{**}	0.64	0.47	$+0.4$	-5.5^{***}	mixed profile
NLD	-0.8	0.48	0.71	$+1.5$	-0.1	null
SWE	-3.2	0.01	0.07	-3.2	-2.8	pre-trends fail
BEL	-5.2^{**}	0.05	0.05	-1.3	-6.6^{*}	pre-trends fail
Non-EU OECD	-1.5	0.24	0.43	-0.7	-1.5	null

Notes. Outcome: indicator that the investor syndicated with a partner from the corridor country in the semester. Early classification (pre-2014), panel 2014–2025, investor and semester fixed effects, clustered on investor country, wild-cluster bootstrap-t with $B = 999$. Pre-trend (i): joint test on $\tau \in \{-5, -4, -3\}$; (ii): joint test additionally including $\tau = -1$. Both are computed by restricted wild-cluster bootstrap. Verdicts are assigned mechanically by the ex-ante grid of Section 5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Eleven of the thirteen corridors return nothing. Two clear the grid: the United Kingdom and Switzerland. We report this rather than tighten the grid until only one survives, and the distinction between them is not made by the static filter but by the

timing, which Figure 2 displays.

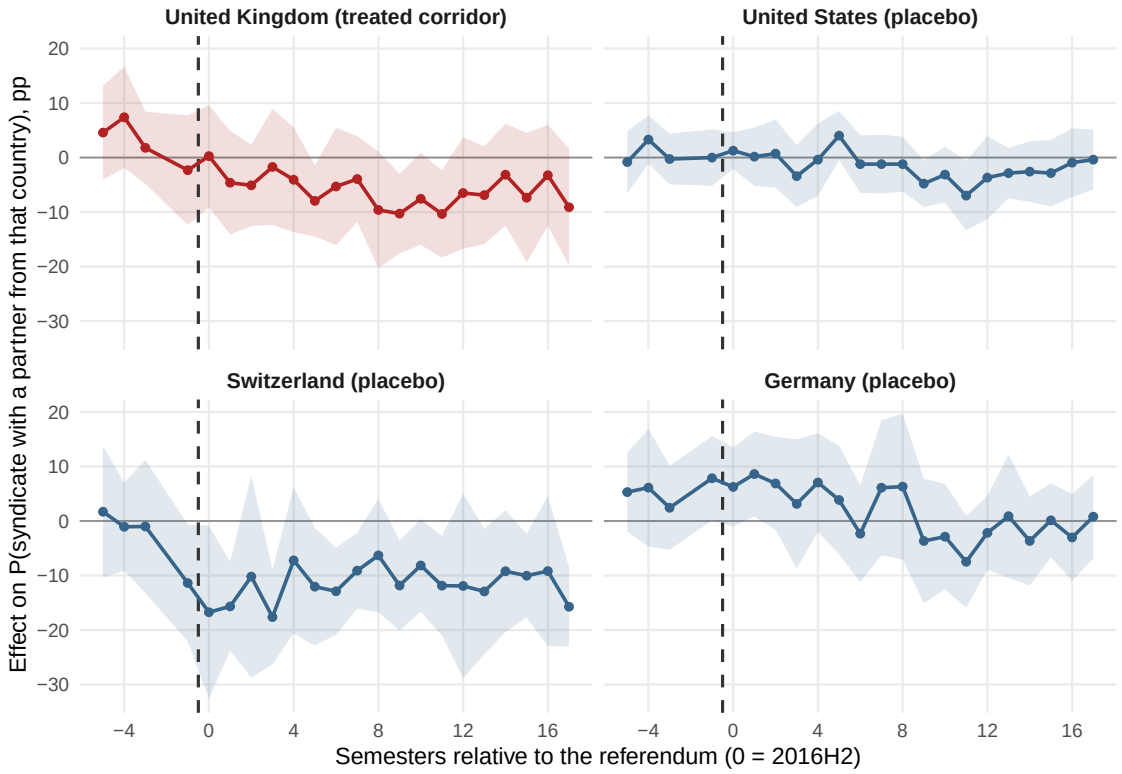


Figure 2: Four corridors, identical axes and identical specification. Only the British corridor declines from the referendum onwards. The Swiss corridor has already collapsed by $\tau = -1$, the semester of the campaign, before the vote, and then plateaus; the American and German corridors move only from 2020. Shaded bands are 95% confidence intervals.

The Swiss corridor breaks at $\tau = -1$, with a coefficient of -11.4 ($t = -2.1$) in the semester of the referendum campaign itself, and is flat thereafter: whatever moved UK–Swiss co-investment was already complete before the vote was counted, and no channel running from a referendum on Union membership can produce a break that precedes it. The German and American corridors move only in the final window. The British corridor is the only one whose decline is dated on the referendum and sustained after it.

We note the asymmetry of what this establishes. The static filter admits two corridors; the timing admits one. A reader unwilling to weigh the timing evidence should conclude that this design identifies a decline in British and Swiss partner rates and cannot separate them, which would leave the paper's claim intact in form but weaker in specificity.

Exposure is a share and its denominator is size: a fixed number of British partnerships is a larger fraction of a smaller book. Highly exposed investors are accordingly smaller

and less active than their less exposed peers, which is why every specification carries the $\text{small}_i \times \text{post}_s$ control of Section 5. Three further checks ask whether the estimate is an artefact of that correlation.

Treated investors are not older. Their first recorded deal comes 1.4 years later than that of controls (2005.0 against 2003.6), and within size stratum the difference vanishes: 2003.2 against 2003.5 among large investors, 2006.3 against 2006.2 among small ones. The age gap is a reflection of the size composition, and the size control absorbs it.

Table 4 reports the rest. Allowing each founding cohort, each quartile of pre-2014 activity, and each cohort-by-size cell its own time path leaves the estimate between -8.2 and -9.7 points. Truncating the sample from below leaves it flat: -8.8 among investors with at least six pre-2014 deals, -9.4 at ten, -8.5 at twenty. It falls to -3.7 only beyond thirty, where sixteen treated investors remain and the interval covers zero. Measured by activity quartile the effect is largest in the third, among investors averaging 1.8 deals per semester, so it is not carried by the least active firms.

Table 4: The first stage under alternative treatments of composition.

Specification	Estimate (pp)	p
Baseline (Table 3)	-8.20	0.000
Age controls		
+ founding year × post	-8.30	0.000
+ above-median age × post	-8.39	0.001
Group-specific time paths		
Semester × founding cohort	-8.15	0.000
Semester × pre-2014 activity quartile	-9.68	0.006
Semester × cohort × size	-8.18	0.000
Truncation from below		
≥ 6 pre-2014 deals	-8.78	0.000
≥ 10	-9.35	0.014
≥ 15	-8.26	0.003
≥ 20	-8.52	0.026

Notes. Outcome, sample and estimator as in Table 3. Founding cohort is the quartile of the year of the investor's first recorded deal; activity quartile is the quartile of pre-2014 deal count. p-values from restricted wild-cluster bootstrap-t on investor country, $B = 999$.

The estimate pools twenty-one countries of origin, and Germany and France together account for 196 of the 448 investors. Removing each country with at least ten investors in turn, and then Germany and France together, leaves it between -7.0 and -11.0 points, significant at the 5% level and with both pre-trend tests passing in all fourteen configurations (Appendix C).

7 The boundaries of the effect

The shift in partner choice is the whole of the effect. This section establishes where it stops, which is what makes it interpretable: it severed no relationship, destroyed no volume, changed nothing about what these investors did, and has no identified counterpart on the British side.

7.1 It did not break relationships

The natural place to look for a divorce is the tie itself. We construct every dyad (i, j) of European investors that co-invested at least once before 2014, label it *persistent* if it shared at least two deals and *occasional* otherwise, and ask whether it is reactivated, that is, shares at least one further deal, in each subsequent semester. The treated dyads are the 1,347 pairs with a British partner; the control dyads are the 7,224 continental–continental pairs.

$$\text{active}_{ds} = \beta_1(\text{persistent}_d \times \text{post}_s) + \beta_2(\text{persistent}_d \times \text{post}_s \times \text{UK}_d) + \beta_3(\text{post}_s \times \text{UK}_d) + \Gamma X_{ds} + \alpha_d + \delta_s + \varepsilon_{ds} \tag{2}$$

β_2 is the Brexit effect net of the natural decay of relationships. It is $+0.30$ percentage points, with $p = 0.77$ and a 95% confidence interval of $[-1.84, +2.51]$ on a semi-annual reactivation base rate of 5.63% (197,133 observations, 29 clusters). We exclude any reduction in the survival of British ties exceeding 1.8 points, one third of the baseline rate.

Meanwhile $\beta_1 = -2.26$ points: persistent ties decay, everywhere, at the same rate regardless of the partner's nationality.

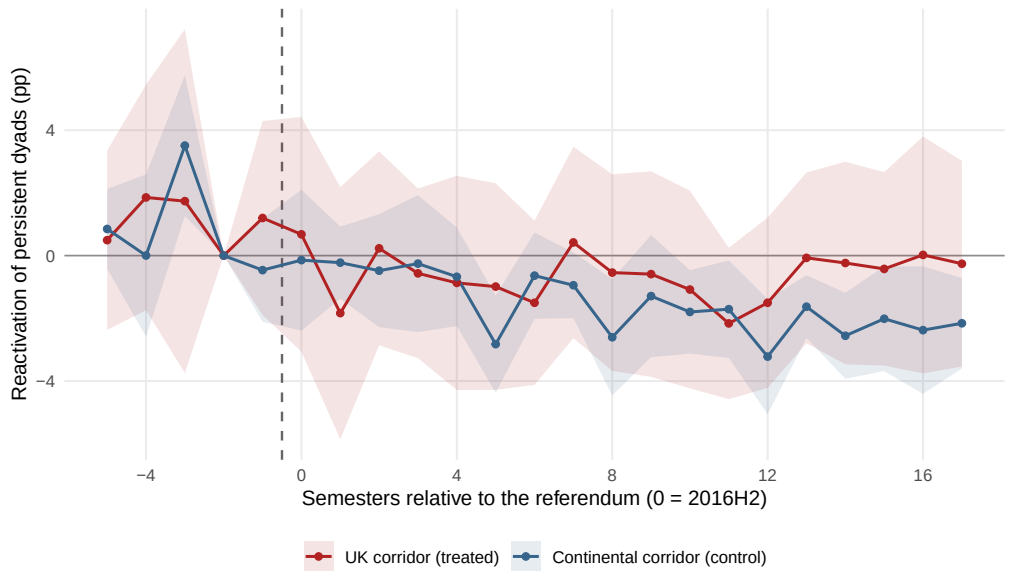


Figure 3: Reactivation of persistent dyads, UK corridor versus continental corridor. Early classification (pre-2014). The curves coincide: British ties do not die faster.

This is the result that gives the paper its title. The investors we study did not sever their British relationships; they stopped adding British partners to new syndicates. What changed is selection, not destruction: avoidance, not divorce.

Classifying dyads on the full pre-referendum window instead of pre-2014 yields -7.1 points, significant at all conventional levels. It also yields -5.1 points for France and -8.0 points for Germany, equally significant. A dyad observed with two or more deals just before the cut-off is, by construction, at a local peak of its activity; it reverts. The correction of Section 5 removes this entirely, and with it the divorce.

7.2 It did not shrink the corridor

If ties did not break individually, did the corridor shrink in aggregate? We build a panel of country-pair corridors. The unit is an unordered pair of European countries, and the outcome is asinh of the number of deals jointly syndicated by investors from both countries in a semester. We compare the eighteen UK–continental corridors against the fifty-five continental–continental corridors. Crucially, the outcome is defined identically for treated and control units.

The post-referendum differential is $+0.03$ log points, with $p = 0.81$ and a 95% confidence interval of $[-0.204, +0.261]$: we bound any effect on the volume of UK–continental syndication within $[-20\%, +26\%]$. Joint pre-trends are flat ($p = 0.87$) and a placebo at 23 June 2014 returns $+0.12$ ($p = 0.33$).

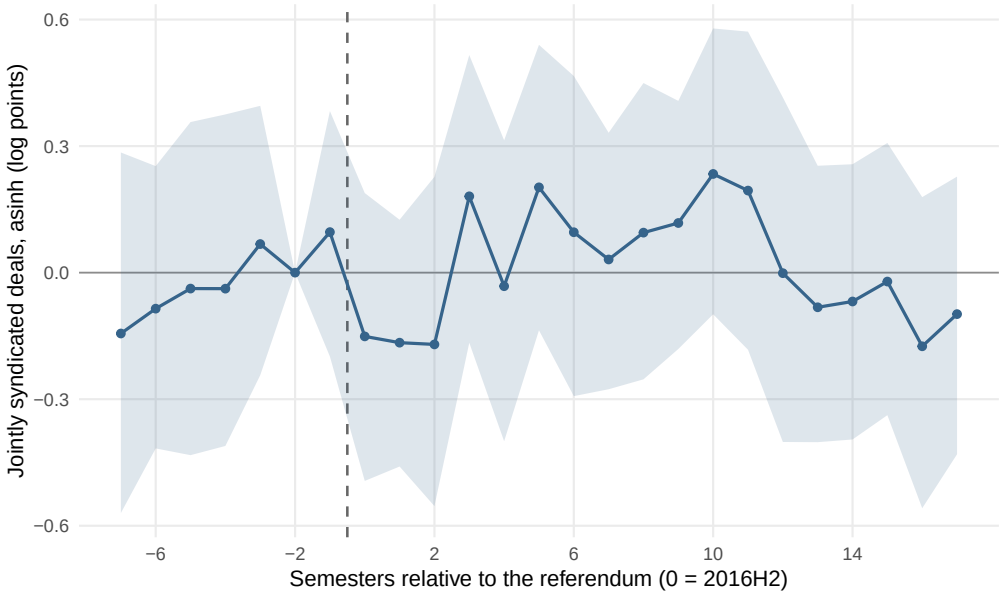


Figure 4: Volume of jointly syndicated deals: UK–continental corridors versus continental–continental corridors. Wild-cluster bootstrap-t, $B = 999$.

The two nulls together are more informative than either alone. Individual British ties survive at the same rate as continental ones, and the aggregate corridor carries the same volume. A compositional shift of 8.2 points that leaves both untouched is a shift in *which* British partners are chosen and how often they are added to new syndicates, not a contraction of the relationship between the two markets.

7.3 It did not change what these investors did

A natural conjecture is that investors reducing their reliance on British partners compensate elsewhere. Comparing small, highly UK-exposed continental VCs to same-size peers never exposed to a British partner returns +6.7 percentage points on the probability of being active ($p < 0.001$) with clean pre-trends. Taken alone, it looks like a finding.

It does not survive the battery. Applying the identical protocol to the same corridors, exposure to American partners produces +7.4 points, to Swiss partners +5.6, to a non-EU OECD aggregate +7.7, all with clean pre-trends and effects in every window, and all clearing the ex-ante grid that the United Kingdom does not. Whatever this coefficient measures, it is not specific to Brexit. It is also mis-timed: in the window immediately following the referendum the UK coefficient is +3.2 points with $p = 0.18$, and the effect appears only in 2020–21 and 2022–25, during the global venture boom.

The conventional control group, investors never exposed to the partner country, is not a group of comparable firms that happened to lack a British partner. Of the 757 small

continental VCs never exposed to a UK investor, 548 (72%) never co-invested with *any* foreign European partner at all. The comparison therefore contrasts networked investors with isolated ones, and networked investors capture a boom better, at any date. Restricting the control group to equally networked VCs halves the estimate to +3.6 points ($p = 0.06$); restricting it to the isolated firms *raises* it to +7.9, which is the diagnostic.

We are careful here: this is not a precise null. The interval admits effects up to +8.2 points, and with 127 treated small VCs we lack the power to exclude a genuine response. Our claim is that the reported adaptation is *not identified*, not that it is zero.

The remaining outcomes are null and precisely so. Deal count, stage composition, the share of deals with continental or American targets, mean round size, follow-on financing and exits within four years: all are reported in Appendix B, all estimated on the same sample with the same design. The geography result is worth stating in the text because it is the one a reader will expect to move: exposed investors did not redirect their deal flow towards continental targets (+2.5 pp, $p = 0.55$), and did not report towards American ones.

7.4 It has no identified counterpart on the British side

The universe of this paper is continental, and that choice requires a defence. The obvious reading is that both ends of a corridor feel a shift in it, and a British panel duly returns a first stage of −6.0 points ($p = 0.039$).

Nothing in the protocol is continental. It holds the universe of investors fixed and varies the partner corridor; setting that universe to the United Kingdom asks of a British VC exactly what Table 3 asks of a continental one. The prediction is in fact sharper on this side: Brexit changed the United Kingdom's relationship with the Union and with nothing else, so EU corridors should move and non-EU corridors should not.

They do not. The continental aggregate reproduces in magnitude but fails both joint pre-trend tests ($p = 0.04$ and $p = 0.03$): the high-exposure group was already converging before the vote. Meanwhile British withdrawal from *American* partners clears the grid outright (−5.0 points with pre-trends at $p = 0.95$ and $p = 0.97$, negative in every window, −7.1 points in 2022–25), and withdrawal from Swiss partners reaches −10.9. France, Germany and the Netherlands are null. The directional prediction is not merely unmet; it is inverted.

We state the implication narrowly. The British panel is small enough that the minimum-cell rule drops seven of fourteen corridors, and the French, German and Dutch nulls rest on forty to fifty treated firms: we cannot exclude genuine British withdrawals of the magnitude they report. The claim is that the British side is *not*

identified by this design, which is why the paper's universe is continental and why we describe an effect on continental partner choice rather than a symmetric rupture.

8 The margin on which the shock operates

A shift in partner choice, dated to a vote, sustained for nine years, unaccompanied by any change in volumes, relationships, activity, or downstream outcomes, and occurring where no rule was ever binding. What produced it?

The design answers one part of that question directly. A semester in which a continental investor syndicates with a British partner is of exactly one of two kinds: either at least one British partner in it is being met for the first time, or every British partner in it has been met before. The two indicators are mutually exclusive and sum to the headline outcome. Running the specification of Section 5 on each therefore returns two coefficients that sum to the headline estimate by construction, with no residual to allocate. We call the first the formation margin and the second the continuation margin, and we subject each to the same protocol and the same thirteen corridors as the headline.

Table 5: The headline effect by margin.

	Estimate (pp)	p	Pre-trends (3 / 4 lags)	Base rate	Verdict
UK-partner rate	−8.20	0.000	0.23 / 0.36	18.7%	durable break
Formation margin	−4.84	0.023	0.15 / 0.14	15.3%	durable break
Continuation margin	−3.36	0.073	0.91 / 0.95	3.4%	not identified

Notes. The two margins are mutually exclusive and exhaust the headline outcome, so the two estimates sum to it exactly. Wild-cluster bootstrap-t p-values ($B = 999$, Rademacher, clustered on investor country). Pre-trend columns report the joint Wald tests on $\tau \in \{-5, -4, -3\}$ and on $\tau \in \{-5, -4, -3, -1\}$. Verdicts are those returned by the reading grid of Section 5, applied unchanged.

Table 5 reports the split, and Figure 5 plots the two margins semester by semester. Of the 8.2 points, 4.8 sit on the formation margin and 3.4 on the continuation margin. The shock is therefore not confined to one margin, and we do not claim that it is.

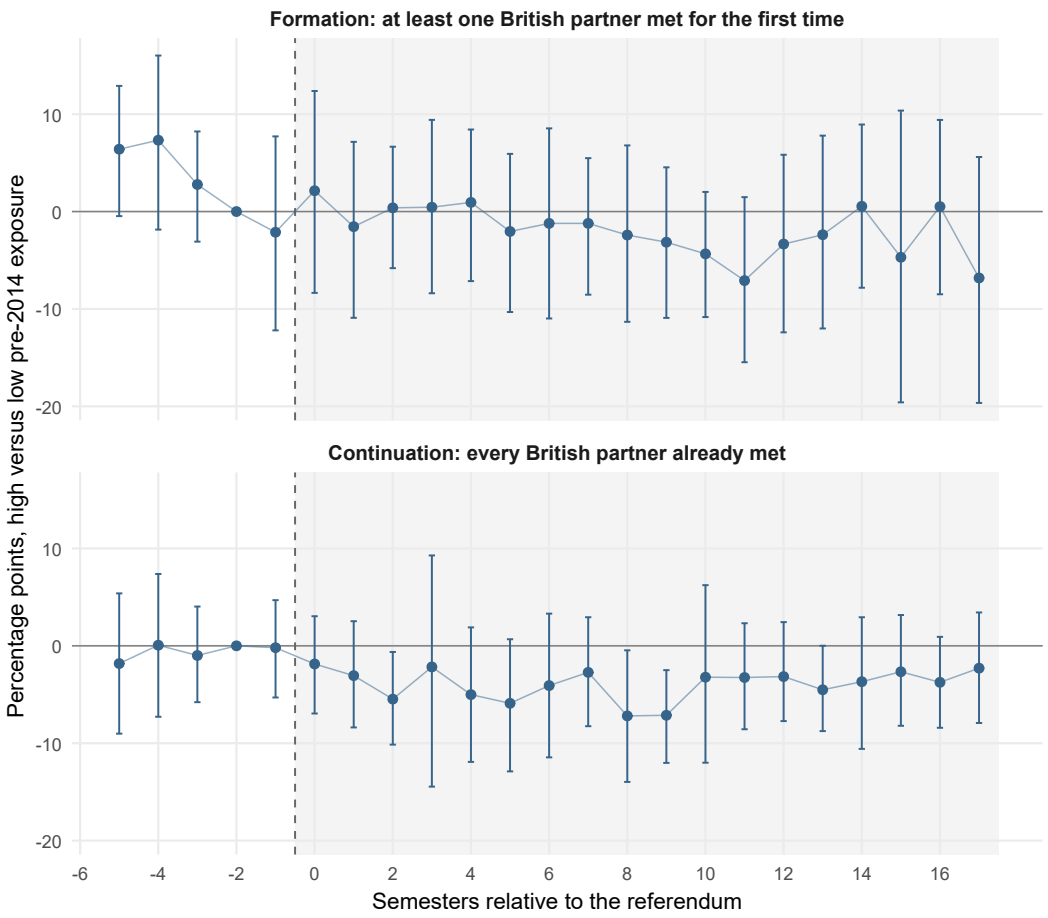


Figure 5: The headline effect decomposed by margin. Upper panel: semesters in which at least one British partner is met for the first time. Lower panel: semesters in which every British partner has been met before. Same specification, sample and fixed effects as Figure 1; wild-cluster bootstrap-t bands ($B = 999$, clustered on investor country). Reference semester: 2015H2.

What separates the two margins is not the size of the estimate but what the falsification protocol returns. On the formation margin the British corridor is the only one of thirteen that survives the grid: a significant static coefficient, pre-trends that pass both joint tests ($p = 0.15$ and $p = 0.14$), and a negative effect in each of the three post-referendum windows, persisting into 2022–25. None of the twelve placebo corridors does. Belgium returns a formation coefficient of -4.1 points significant at $p < 0.001$ and fails both pre-trend tests at $p < 0.01$. Korea passes both pre-trend tests but its third-window effect does not reach significance ($p = 0.11$). The remaining ten are null.

On the continuation margin no corridor survives the grid, the United Kingdom included. Its estimate is -3.4 points on a base rate of 3.4%, with pre-trends that are exceptionally clean ($p = 0.91$ and $p = 0.95$), but the coefficient falls short of conven-

tional significance ($p = 0.073$) and lies below the minimum detectable effect of that specification, 4.7 points. We therefore do not read it as a zero, and the same caution applies here that we applied to our own British estimate in Appendix A. What we read is the comparison: the formation margin separates the British corridor sharply from twelve placebos, and the continuation margin does not separate it from them at all.

One feature of the formation estimate should be raised rather than left to a reader. Its first-window coefficient is -3.0 points and does not reach significance ($p = 0.14$), where the headline reaches it comfortably (-6.4, $p = 0.015$). The sign is right and the effect is present in all three windows, but the formation margin is estimated on a thinner outcome and carries correspondingly less precision early.

A second feature belongs here, and it bounds what the split can be said to establish. The continuation margin is defined as a semester in which every British partner is already known, so a semester in which the investor meets one new British partner and re-engages two known ones is counted entirely as formation. That convention is what makes the two margins exhaust the headline outcome, and it is also what holds the continuation base rate down to 3.4%. Measuring the same split in counts of distinct partners rather than in semesters removes the convention and reverses the shares, the continuation margin then carrying the larger part. We do not report those estimates as results: all three count outcomes fail both joint pre-trend tests, a count following the investor's level of activity, which diverges across exposure groups before the vote where the bounded indicator does not. The split we report is the one this design identifies, and we cannot show it invariant to the definition of the outcome.

Three classes of explanation were already eliminated. It is not compliance: no rule governing syndication changed at any date in our sample (Section 2), and the response precedes the only legal change by four and a half years. It is not a change in the opportunity set: restricting the sample to portfolio companies that already existed before June 2016 leaves the pattern intact. And it is not a general retreat from foreign partners: the same investors' exposure to American, Canadian, Japanese, French, German, Dutch and non-EU OECD partners is unchanged.

Nor can we say where the displaced partnerships went. Among new partners, the British share falls by five points ($p = 0.009$ in 2018–19) and the continental share rises by an almost exactly offsetting amount, but only the first estimate is identified: the second disappears when the ratio is restricted to semesters in which it is measured on at least three new partners, a restriction the British decline survives (-4.7 points, $p = 0.024$). The domestic share – partners from the investor's own country – rises in the restricted sample, and we submitted it to the thirteen-corridor protocol rather than report it as a destination. It does not survive: the British static coefficient is insignificant ($+2.0$, $p = 0.13$) and its pre-trends fail both joint tests, while the only corridor returning a sustained response is the German one ($+4.5$, $p < 0.001$, pre-trends at 0.55 and 0.66, an effect in all three windows). Retreat towards domestic partners is a general feature of the period, and it is not identified on the British

corridor. Five points of partner choice spread over the rest of the market leave no destination this design can attribute.

The margin constrains two more.

The first is an upstream constraint. European public institutions are substantial limited partners in continental venture funds, and a tightening of their geographic mandates would transmit a quasi-regulatory constraint into the syndication decision without appearing in any rule governing syndication itself. Three things bound it.

The margin bounds it first. A mandate of that kind binds on new investments, and a follow-on round alongside a British investor already in the syndicate is a new investment. It should therefore bind on both margins. It is on the formation margin alone that the British corridor separates from the placebos.

Timing bounds it second, and more sharply than the margin does. The mandates that govern European public limited partners restrict the domicile of portfolio companies and the establishment of the fund; we are aware of none that restricts the nationality of a co-investor, and a continental fund could syndicate with a British one throughout. What such a constraint can plausibly mean here is therefore not a rule but a belief about a rule: caution over eligibility, an anticipated complication in reporting, a compliance reflex in the absence of any requirement. A belief of that kind has an expiry date. By 31 December 2020 the separation was complete, the mandates were whatever they had become, and the absence of any eligibility consequence for syndication was observable. The response does not attenuate: -6.4 points in H2 2016–2019, -10.6 in 2020–21, -8.3 in 2022–25. A mistaken belief about eligibility that survives five years past its own refutation is a demanding assumption.

The third bound is that this channel, unlike the two we cannot separate, makes a cross-sectional prediction. A belief about public mandates has a carrier: it has no reason to exist in funds that are not backed by public money, where neither the coordination cost of a new relationship nor the salience of a redrawn boundary has any reason to vary. We attempt that test in Section 8.1 and it does not identify. The version of the hypothesis written specifically on the opening of new relationships would in any case require fund-level LP data to eliminate outright.

The second is anticipation of a bilateral friction. Investors may have priced frictions expected to arrive with legal separation: on fundraising, on establishment, on the mobility of personnel. A friction of that kind attaches to the partner, not to the acquaintance: a 2017 syndicate formed with a British investor already known extends past 2020 exactly as one formed with a British investor newly met. Anticipation of this form predicts a response on both margins in the same proportion, and the protocol identifies one on only one.

Two channels predict a response concentrated where relationships are opened. Syndication binds partners to years of joint governance (board seats, follow-on decisions, exit timing), and uncertainty about the future of a bilateral relationship raises the

cost of entering it today even when the present cost is unchanged; the coordination already established in an existing tie is not re-incurred. Alternatively, a vote redraws an institutional boundary and makes it salient, and networks may realign on that boundary through attention, reputation and expectation at the point where partners are matched, which is the channel the trust literature (Guiso et al., 2009; Bottazzi et al., 2016) would predict. Both are consistent with the margin we identify.

The timing separates them from a narrower version of themselves. The coordination channel runs on uncertainty about the future of the relationship, so it is strongest while that future is open: the referendum opened it in June 2016, the votes of 2019 were its peak, and the agreement of December 2020 closed it. Estimated over four windows the effect runs the other way: -5.4 points in H2 2016–2017 ($p = 0.064$), -7.1 in 2018–19 ($p = 0.008$), -10.1 in 2020–21 ($p < 0.001$) and -8.9 in 2022–25 ($p < 0.001$). The formation margin is at its lowest in the final window (-6.2 points, $p = 0.032$). Hysteresis reconciles a permanent stock with a temporary shock, since a tie not formed for four years is not formed later, but it predicts that the flow of new ties returns to normal once the regime is known, and that flow does not return. What the coordination channel can mean here is therefore a cost of entering a cross-Channel relationship that is permanent rather than one that expires with the uncertainty, which is the same statement the boundary channel makes. Separating the two would require data on expectations or on the terms of individual syndicate negotiations, which we do not have.

8.1 A test of the upstream channel that does not identify

The cross-sectional prediction is the one a reader will propose, so we report what happens when it is run. Crunchbase records investors and not their limited partners, so the carrier we can observe is the ecosystem rather than the cap table: we flag the seventy-three European public investment entities that appear in our data (the European Investment Bank, Bpifrance, KfW and the High-Tech Gründerfonds, Almi Invest, Enterprise Ireland, Bayern Kapital, CDP Venture Capital, Tesi, Finnvera, PMV, NRW.BANK, Invest-NL, Portugal Ventures and their peers) and classify a continental venture capitalist as publicly connected if it syndicated alongside any of them before 2014. The classification is pre-determined on the same window as exposure. Twelve investors in the British corridor are themselves public entities and are set aside rather than assigned to either side.

The cells are adequate: among the highly exposed, 157 privately connected against 59 publicly connected; among the weakly exposed, 125 against 95. All four exceed the minimum cell we impose everywhere else.

The estimates cannot be read, for two reasons that are independent of each other.

Both subsamples fail both joint pre-trend tests ($p = 0.01$ and $p = 0.02$ for the privately connected, $p = 0.00$ and $p = 0.00$ for the publicly connected), where the pooled panel

passes them comfortably ($p = 0.23$ and $p = 0.36$). The two subgroups diverge before the referendum in directions that offset once pooled. Parallel pre-trends in the pooled panel is the assumption this design requires, and it holds; it does not follow that it holds in an arbitrary partition of that panel, and here it does not. Neither the triple interaction nor either subsample coefficient is identified, and we report none of them as an estimate.

The cut is also not silent where it should be. Applied unchanged to the American corridor, the same triple interaction returns +9.7 points ($p < 0.001$), with the publicly connected subgroup significantly positive on pre-trends that are entirely clean. Public backing separates funds whose trajectories differ for reasons that have nothing to do with any referendum (domestic orientation, stage, the growth path of the vehicles that carry public money), and a partition that produces a significant differential in a placebo corridor cannot be used to attribute one in ours.

What bounds the upstream channel is therefore the timing argument above and not this partition. We report the attempt because the partition is the obvious one, and a reader who constructs it should know in advance what it returns.

9 Discussion

9.1 Integration rests on behaviour, not only on rules

The legal architecture of European financial integration was, on this margin, entirely undisturbed. A continental investor's capacity to syndicate with a British one was never conferred by membership and was never withdrawn. If integration on this margin were sustained by the framework, nothing should have happened.

Something did, and it was large: a forty-four percent relative decline in the rate at which the most exposed continental investors take a British partner, beginning at the vote and still present nine years later.

The implication runs in both directions, and the second is the uncomfortable one. Dismantling the legal framework was not necessary to disintegrate the network: an expectation sufficed. And symmetrically, preserving or restoring the legal framework will not be sufficient to reintegrate it. For the Capital Markets Union agenda this is a caution rather than a prescription: instruments that operate on rules address the part of integration that, here, was never the binding one.

We would not extend this beyond the margin we measure. Volumes did not move; relationships did not break. What we document is that the *selection* of partners, the margin that determines an investor's future opportunity set (Hochberg et al., 2007, 2010), responded to an institutional signal that changed no rule.

9.2 An unexpected effect, and a bounded one

Two things about this result are worth holding together, because each disciplines the other.

The first is that it should not have happened. No rule governed the margin it moves. The channels one would reach for have each been given a test and each returns nothing: fundraising and the composition of capital (Section 8.1), the activity of the exposed investors (Section 7.3), the volume of the corridor (Section 7.2), the survival of established ties (Section 7.1), and the British side of the same pairs (Section 7.4). What remains is a change in whom these investors agree to meet: the formation margin breaks durably and clears the protocol, while the continuation margin returns a null. Investors did not withdraw from partners they had; they stopped taking on new ones. An expectation was enough to move that, and no instrument had to change hands.

The second is that the same battery bounds what the result licenses. Deal counts, stages, round sizes, follow-on financing and exits are all precise nulls. Aggregate volume in the UK–continental corridor tracks the rest of Europe. A network of partner choices was reorganised over nine years without the markets it operates in registering the reorganisation. Whatever dependence between these two financial systems the withdrawal disturbed, it was not load-bearing for the outcomes those systems produce.

That second reading is the one we would press, because it cuts against the direction in which this literature usually errs. A market-wide institutional shock can move behaviour that no rule reached. That is the first finding, and it is a caution against reading integration off the legal framework. But the size of the cross-border dependence such a shock threatens cannot be inferred from the fact that it disturbed something. Here it disturbed a selection margin and stopped there. The dependence was real, it was relational, and on the evidence of the outcomes it was smaller than the withdrawal debate assumed.

9.3 A note on identification under market-wide shocks

Our design choices were forced by a feature of the setting worth stating for other settings that share it. Brexit reaches every European investor on the same date, so there is no untreated unit. The econometric literature on difference-in-differences has focused on biases arising from heterogeneous effects and staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; Roth et al., 2023), all of which presume that some units are, for a while, untreated. Here none are.

The practical consequence is that the aggregate time trend is not absorbed. It loads onto the treatment indicator and produces coefficients that are large, correctly signed,

significant, and spurious. We encountered three in the course of this project: a dyadic divorce of -7.1 points that is regression to the mean and reproduces for France and Germany; an activity expansion that reproduces for American and Swiss exposure; and a symmetric British rupture that dissolves once exposure is classified outside the window in which pre-trends are then tested. None was removed by a better estimator. Each was removed by a contemporaneous counterfactual: a placebo corridor, a control group matched on network position, the same grid pointed at our own second panel.

One further observation may be useful to practitioners. In this project the joint pre-trend test was consistently the weaker safeguard and the date placebo the stronger one. Several results passed pre-trend tests comfortably, including some with p-values above 0.9, and were then eliminated by a placebo that placed the treatment date in 2014, where the design returned effects of the same magnitude as the real one. Where the two disagreed, the placebo was right. We report this because the profession's default order of operations is the reverse.

The aggregate trend is visible in our own control groups, and we report what it does because it is the quantity a reader should want to see before believing any of this. Among continental investors with low pre-2014 exposure to a given partner country, the probability of syndicating with a partner from that country rises after 2016 in every one of the thirteen corridors: by 131% of its own pre-vote level for Korea, 71% for Belgium, 54% for the Netherlands, 3.5% for France, with a median of 47%. The British corridor's control group rises by 20.5%, ninth of the thirteen. The expansion against which our treated investors fail to move is therefore not specific to Britain, and is of exactly the kind that semester fixed effects are there to absorb.

Nor is it an artefact of the source. Crunchbase records 1.85 investors per round over 2014–15 and 1.88 over 2017–19, a change of 1.2%, and the share of rounds recording at least one British investor stays within a band of nine to twelve percent across the whole period without trend. Recording that moves by one percent cannot generate group differences of the size we estimate. Investors per round do rise later, to 2.37 in 2021, which is one further reason to rest nothing on the 2020–21 window.

A last observation about how the effect presents itself. Raw pre-post changes in the treated groups do not single out the United Kingdom: Switzerland falls by 2.5 points, Germany by 2.0, France by 1.8, Korea by 1.5, the United Kingdom by 1.4. This is what a contrast that ignores pre-trends, fixed effects, the size control and persistence should be expected to show, and those are precisely what separate the corridors: Switzerland breaks during the campaign, Germany arrives late, France fails both pre-trend tests. The result is a property of the falsification grid rather than of the raw table, and we would rather say so than leave it to be discovered.

10 Conclusion

We summarise. Nothing in the withdrawal of the United Kingdom from the European Union restricted co-investment, at any date. Continental venture capitalists whose deal flow had been built around British partners nonetheless reduced the share of their syndicates including one by 8.2 percentage points ($p < 0.001$; 95% CI $[-13.1, -3.2]$) against a base rate of 18.7%. The decline begins at the referendum, four and a half years before the single market ceased to apply to the United Kingdom, and is still present in 2022–2025.

The effect is confined to that margin. Established British ties survived at the rate of comparable continental ones (+0.3 pp, $p = 0.77$; we exclude reductions beyond a third of the baseline). The aggregate corridor carried the same volume, within $[-20\%, +26\%]$. Deal counts, stages, target geography, round sizes, follow-on financing and exits are null. Whatever these investors lost in British partners, they did not visibly replace, redirect, or suffer from.

Among thirteen partner corridors subjected to an identical protocol, eleven return nothing and two clear the grid; of those two, only the British corridor declines from the referendum onwards, the Swiss having already broken during the campaign that preceded it.

We identify the margin, not the channel. The effect splits exactly into -4.8 points on semesters that open a new British relationship and -3.4 on semesters that continue an existing one; under the same protocol the British corridor is the only one of thirteen to survive on the first margin, and no corridor survives on the second. An anticipated bilateral friction and a geographic mandate transmitted upstream by public limited partners both predict a response on the two margins. What remains are the channels that operate where partners are matched, the coordination cost of a new bilateral relationship and the realignment of a salient institutional boundary, and separating those two requires data on expectations or on the terms of individual syndicate negotiations that we do not have.

What we take from this is a fact about how integration is held together. The rules that governed syndication between British and continental investors did not change, and the syndication changed anyway. A framework left intact did not preserve the behaviour it is usually credited with sustaining, which suggests, for anyone whose instruments operate on rules, that the rules were not the binding constraint.

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A Precision: what the nulls exclude

A null result is informative only if it is precise. Table 6 reports, for each headline estimate, the wild-cluster bootstrap confidence interval obtained by inverting the bootstrap distribution, the base rate of the outcome in the control group before the referendum, and the minimum detectable effect at 80% power.

Table 6: Precision of the headline estimates.

		Estimate	95% CI	Base rate	MDE
Effect	UK-partner rate (pp)	−8.20	[−13.11, −3.15]	18.7%	6.64
Effect	formation margin (pp)	−4.84	[−9.24, −0.18]	15.3%	6.20
Not identified	continuation margin (pp)	−3.36	[−7.16, +0.24]	3.4%	4.73
Null	Dyad survival, triple diff. (pp)	+0.30	[−1.84, +2.51]	5.6%	2.62
Null	Corridor volume (log pts)	+0.03	[−0.20, +0.26]	7.5 deals	0.32
Null	Activity, window $[0, 6]$ (pp)	+3.18	[−1.47, +8.16]	26.0%	6.17
Not identified	British side, first stage (pp)	−6.01	[−11.52, −0.33]	22.5%	8.51

Notes. Confidence intervals from inversion of the wild-cluster bootstrap-t distribution (Rademacher, equal-tailed, $B = 999$). Base rate: mean of the outcome among control units before the referendum. MDE: minimum detectable effect at 80% power and 5% size.

The corridor null is decisive: it bounds any effect on the volume of UK–continental syndication within $[-20\%, +26\%]$. Increases in cross-border syndication of the order of 80% that have been read off the post-2016 data lie far outside it.

The dyad null is informative: we exclude any reduction in the semi-annual survival of British ties exceeding 1.8 percentage points, one third of the 5.6% baseline.

The activity null is *not* precise, and we rest nothing on it. Its interval admits effects up to +8.2 points against a base rate of 26%. Our claim about activity rests on the falsification battery, not on precision.

The continuation margin is in the same position, and we treat it the same way. Its point estimate, −3.4 points, lies below the 4.7 points that specification can detect at 80% power, and its interval crosses zero. We do not read it as a zero, and nothing in Section 8 requires that we do: the claim there is that the formation margin separates the British corridor from twelve placebo corridors under the protocol and the continuation margin does not separate it from them, which is a statement about the falsification battery rather than about the precision of either estimate.

The British estimate is not a null at all: its interval excludes zero. It is disqualified by pre-trends and by placebo corridors, and there is an independent warning in this table. The minimum effect that panel can detect with 80% power is 8.5 points, larger than the 6.0 points it reports. A design that returns a significant coefficient below its

own minimum detectable effect is drawing from the tail in which significant estimates are, mechanically, the overstated ones.

B The remaining outcomes

We list every outcome we examined, so that the selection of reported results can be verified rather than trusted. All are estimated on the same sample as the main result (continental VCs, all UK-exposed, high versus low), with early classification, investor and semester fixed effects, a small $\times$ post control, and wild-cluster bootstrap inference.

Table 7: All secondary outcomes.

Outcome	Estimate	95% CI	p	Pre-trends
Deal count (asinh)	-0.03	$[-0.15, +0.08]$	0.52	0.53
Early-stage share (pp)	-0.83	$[-8.30, +8.08]$	0.81	0.20
Continental-target share (pp)	$+2.47$	$[-8.09, +13.38]$	0.55	0.24
Follow-on round within 4 years (pp)	$+0.16$	$[-8.96, +8.96]$	0.96	0.83
Exit, M&A or IPO, within 4 years (pp)	$+4.72$	$[-3.00, +13.35]$	0.16	0.13
Not identified (fails the pre-trend test):				
Mean round size (asinh)	-0.09	$[-0.23, +0.05]$	0.21	0.00

Notes. Pre-trends: joint restricted wild-cluster bootstrap test on $\tau \in \{-5, -4, -3\}$. Mean round size fails it and we therefore report no estimate for it: the groups were already diverging before the referendum, and nothing is identified.

Every outcome we can identify is null. Exposed investors did not do more or fewer deals, did not move to earlier or later stages, did not redirect their deal flow toward continental targets, and their portfolio companies were neither less likely to raise again nor less likely to exit. This is what confines the effect to the relational margin, and it is why we describe the finding as a change in partner choice rather than as a change in behaviour more broadly.

C No single country carries the effect

The estimate pools twenty-one countries of origin, unevenly: Germany and France account for 196 of the 448 investors. This appendix reports it with each country of origin removed in turn, on the sample and specification of the British row of Table 3.

Table 8: Leave-one-out by the investor's country of origin.

Sample	Estimate (pp)	p	Pre-trends (i)	Pre-trends (ii)	Investors high / low
All countries	−8.20	0.000	0.52	0.70	216 / 232
Excluding Germany	−8.61	0.019	0.65	0.81	165 / 178
Excluding France	−9.87	0.000	0.76	0.87	182 / 175
Excluding Switzerland	−8.68	0.005	0.73	0.88	191 / 208
Excluding the Netherlands	−7.25	0.000	0.75	0.91	191 / 214
Excluding Spain	−8.85	0.000	0.38	0.50	205 / 219
Excluding Belgium	−8.19	0.000	0.53	0.66	204 / 222
Excluding Sweden	−8.52	0.000	0.54	0.70	204 / 222
Excluding Denmark	−8.10	0.000	0.32	0.31	206 / 223
Excluding Finland	−8.29	0.000	0.55	0.71	209 / 223
Excluding Norway	−8.41	0.000	0.42	0.58	209 / 224
Excluding Luxembourg	−7.02	0.000	0.64	0.79	209 / 226
Excluding Italy	−8.09	0.000	0.53	0.70	212 / 226
Excluding Germany and France	−10.95	0.000	0.63	0.83	131 / 121

Notes. Outcome, sample and specification as in the British row of Table 3: continental investors with at least two pre-2014 deals and one pre-2014 British co-investor, high versus low pre-2014 exposure, investor and semester fixed effects, a small × post control. Wild-cluster bootstrap-t p-values ($B = 999$, Rademacher, clustered on investor country). Pre-trend (i): joint test on $\tau \in \{-5, -4, -3\}$; (ii): joint test additionally including $\tau = -1$; both by restricted wild-cluster bootstrap. Countries with fewer than ten investors in the sample are not removed individually.

The estimate ranges from −7.02 to −10.95 points, and no configuration fails either pre-trend test. Removing Luxembourg or the Netherlands shrinks it the most, to −7.02 and −7.25. Among single removals, removing France enlarges it the most, to −9.87: France has the lowest share of high-exposure investors among the twelve countries, 37% against 58% in the Netherlands. Removing Germany and France together, which leaves 252 of the 448 investors, returns −10.95.

D The pre-existing opportunity set

The shift in partner choice could in principle reflect a change in the population of companies available to be funded rather than a change in investor behaviour: if fewer British-linked ventures were founded after 2016, exposed investors would take fewer British partners without choosing to.

We fix the opportunity set. Restricting the panel to portfolio companies whose first funding round predates June 2016 (firms that existed, and were fundable, before the referendum), the activity result is unchanged (+5.8 points, $p = 0.002$, pre-trends $p = 0.80$), and a 2014 date placebo on the same fixed pool returns +0.3 points

($p = 0.87$). The pattern is therefore not produced by the entry of new firms into the sample.

We report this as a boundary on the interpretation rather than as support for the activity result, which Section 7.3 declines to identify for independent reasons.